Five rules to evaluate the optic disc and retinal nerve fiber layer for glaucoma

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A systematic approach for the examination of the optic disc and retinal nerve fiber layer is described that will aid in the detection of glaucoma. This approach encompasses 5 rules: evaluation of optic disc size, neuroretinal rim size and shape, retinal nerve fiber layer, presence of parapapillary atrophy, and presence of retinal or optic disc hemorrhages. A systematic process enhances the ability to detect glaucomatous damage as well as the detection of progression, and facilitates appropriate management.

Key Words: Glaucoma, optic nerve, optic disc, retinal nerve fiber layer, optic disc hemorrhages

Fingeret M, Medeiros FA, Susanna Jr R, Weinreb RN. Five rules to evaluate the optic disc and retinal nerve fiber layer for glaucoma. Optometry 2005;76:661-8.

The evaluation of the optic nerve and retinal nerve fiber layer (RNFL) is essential to the recognition of glaucomatous damage. An optic nerve or RNFL abnormality is often, but not always, the first sign of glaucomatous damage.[1,2] In the earliest stages of the disease, optic nerve and RNFL damage may be present, while standard automated perimetry is still within normal limits.[3-6] Early glaucomatous damage can be difficult to detect, requiring careful observation of the optic nerve and RNFL. Optic disc photography or optic nerve and RNFL imaging should be performed at the initial visit and yearly thereafter to document the optic nerve and RNFL status. In situations in which stability is in question, photography and imaging may be done at earlier intervals.

Recent studies have found the difficulty clinicians have in following guidelines proposed by professional organizations.[7,8] These guidelines recommend documentation of the optic disc appearance at the time of diagnosis and at periodic intervals during followup. In one study utilizing a chart review, 193 primary open-angle glaucoma (POAG) patients were followed up in 8 private practices in the Los Angeles area for at least 2 years.[8] Almost all patients had a photograph or drawing at the initial examination, but, at the final followup visit, 33.2% had not had an optic nerve drawing or photograph taken within the previous 2 years. Another 37.8% had not had optic disc photography since the initial examination. A more recent chart review evaluated records from 395 POAG patients in 6 managed care plans.[7] Only 53% had optic disc photographs or drawings at the initial examination.

Although several textbooks and articles describe the characteristic signs of glaucomatous damage to the optic disc,

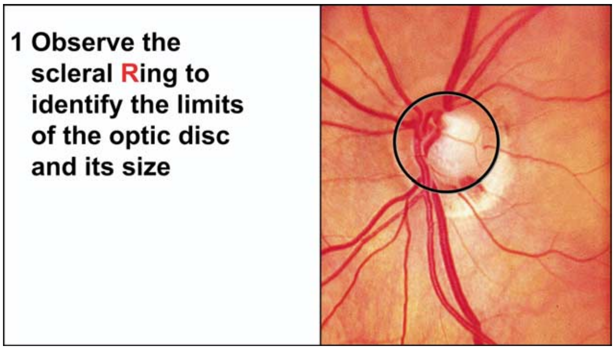

Figure 1 The first rule for the assessment of the optic disc is the observation of the scleral ring and assessment of optic disc size.

Figure 2 The shape of the optic disc is oval, usually slightly greater vertically than horizontally.

Figure 3 The optic disc size varies among individuals, with cup size correlating with the size of the optic disc. These examples are from 4 individuals with different optic disc sizes. The largest is in the top left, followed by top right, lower left, and lower right. Note how the cup size correlates with the disc size, except for the picture in the lower left in which the person has glaucoma with a wedge RNFL defect and large cup.

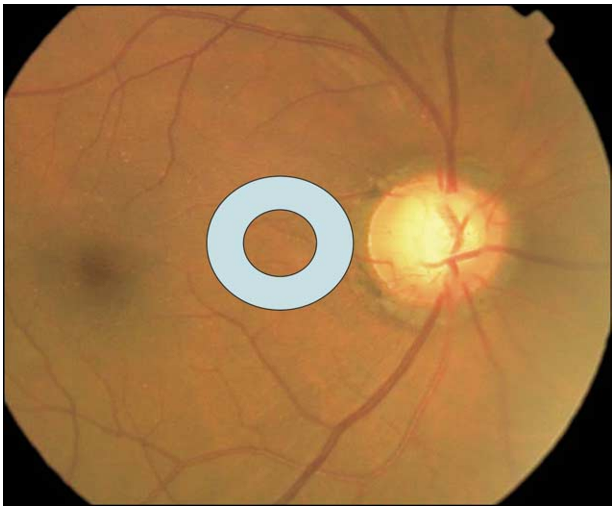

Figure 4 The optic disc size can be evaluated using the small spot of light from the direct ophthalmoscope. In this example of an average size disc, the spot approximates the size of the optic disc.

no systematic approach for optic disc examination in glaucoma has been widely disseminated.[9,10] When examining a patient who either has established glaucoma or is suspected of having the disease, a systematic approach to optic disc and RNFL examination is necessary so that glaucomatous optic neuropathy is not overlooked. A thorough optic nerve examination should be used along with perimetry to diagnose glaucoma and to assess disease severity. Staging the disease and consideration of risk factors for glaucoma progression enables the clinician to establish a target intraocular pressure. The structural assessment (optic nerve and RNFL) and functional evaluation (perimetry) are used together to monitor for change over time as well restage the patient's condition.[1]

In this report, we describe a systematic approach for the evaluation of the optic disc and RNFL in glaucoma that can be incorporated easily into clinical practice. This approach was conceived originally by 3 of the authors and published as a PowerPoint monograph entitled FORGE (Focusing Ophthalmology on Reframing Glaucoma Evaluation) that was sponsored by Allergan, Inc.

Methodology

The five rules (5Rs) for the assessment of the optic disc in glaucoma include:

1. Observe the scleral Ring to identify the limits of the optic disc and evaluate its size.

Figure 5 Certain optic discs, such as those with high myopia, can be difficult to evaluate.

Figure 6 In this example of a person with a tilted disk, the disc margin is difficult to identify.

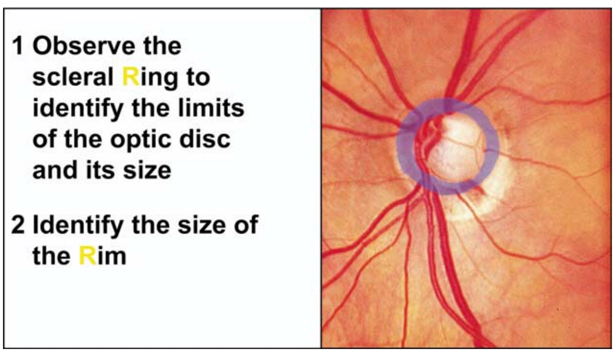

Figure 7 The second rule for the assessment of the optic disc is to identify the size of the neuroretinal rim.

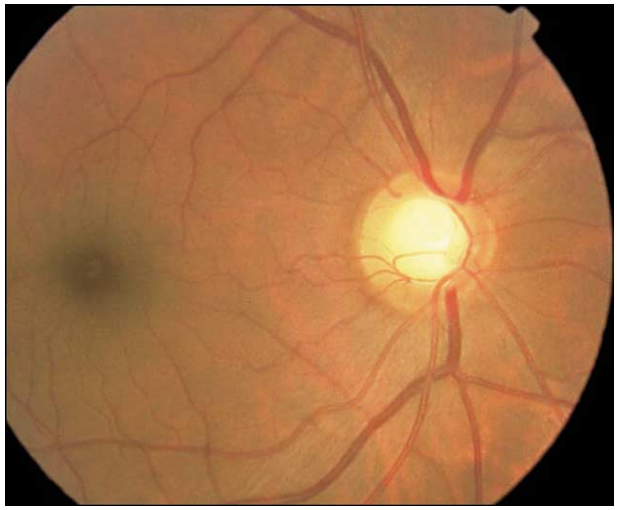

Figure 8 The ISNT rule states that in a healthy optic disc, the widest rim tissue is found inferiorly, then superiorly, nasally, with the temporal rim being the thinnest. This gives rise to a cup shape that is often a horizontal oval.

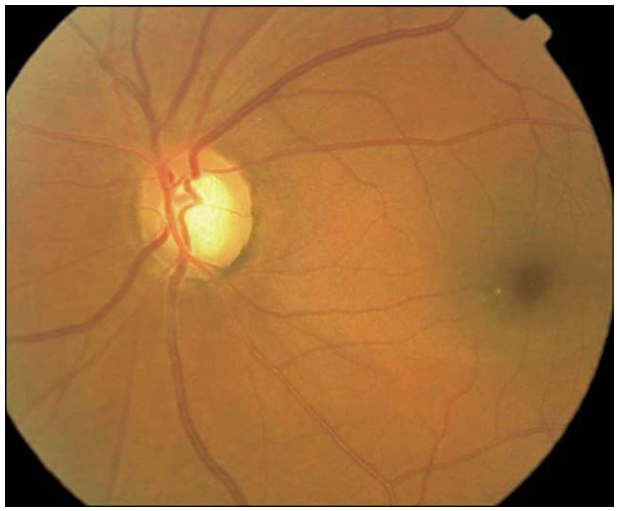

Figure 9 This example shows an optic disc in which the ISNT rule is not obeyed, as seen by the thin rim inferiorly.

2. Identify the size of the Rim.
3. Examine the Retinal nerve fiber layer.
4. Examine the Region outside the optic disc for parapapillary atrophy.
5. Watch for Retinal and optic disc hemorrhages.

Rule 1 recognizes that the first step in evaluating the optic disc for glaucoma is evaluating its size (*see* Figures 1 and 2). Optic discs vary considerably in size, with the size of the optic disc cup correlating with the size of the optic disc.[11,12] In healthy subjects, a small disc (vertical disc diameter $<$ 1.5 mm) will have a small cup, whereas a large disc (vertical disc diameter $>$ 2.2 mm) will have a large cup (*see* Figure 3). Therefore, large

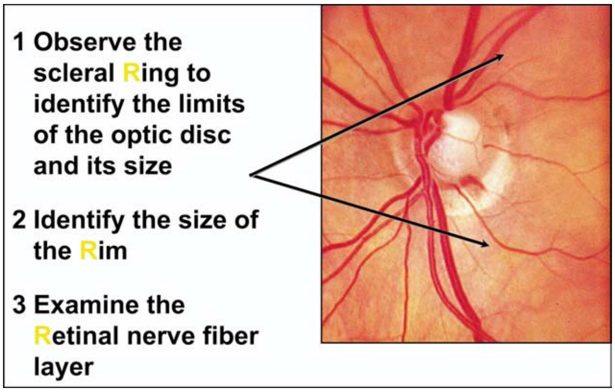

Figure 10 The third rule for the assessment of the optic disc is to examine the retinal nerve fiber layer. The normal RNFL has bright striations in the superior temporal and inferior temporal regions of the optic disc; areas in which the RNFL is thicker. As the RNFL is diffusely lost, the striations disappear, and the borders of small parapapillary retinal blood vessels become more clearly visible.

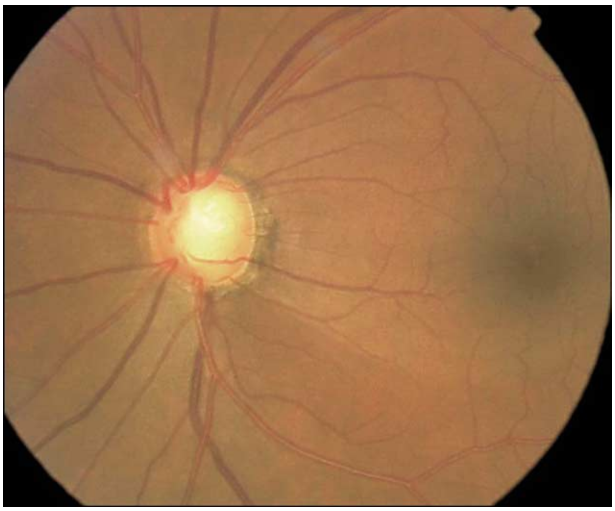

Figure 11 A common way RNFL damage is identified is by the presence of localized loss, as seen by a wedge-shape dark defect at 5 o'clock.

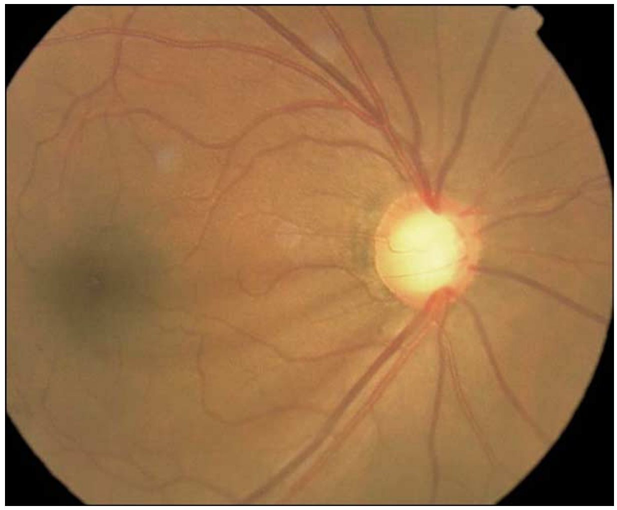

Figure 12 Several small bands of RNFL dropout are seen between 7 and 9 o'clock that extends back to the optic nerve.

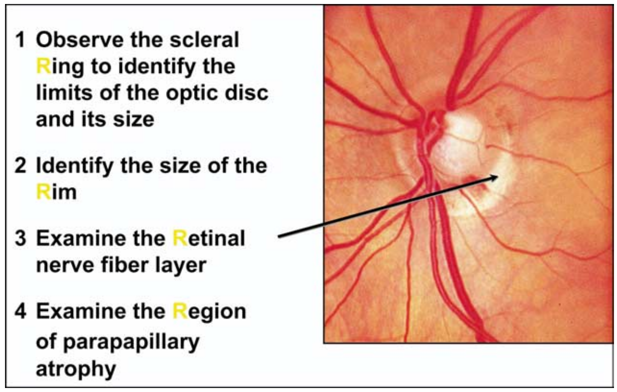

Figure 13 The fourth rule for the assessment of the optic disc is to examine the region adjacent to the optic disc for the presence of parapapillary atrophy.

optic discs in healthy individuals tend to have large cups, which can lead to an erroneous diagnosis of glaucoma. However, small cups can be glaucomatous in small discs.[13,14] There are several methods to evaluate optic disc size. Sophisticated optic nerve imaging techniques such as confocal scanning laser ophthalmoscopy and optical coherence tomography can provide precise measurements. In clinical practice, however, the clinician can assess the optic disc size by using a direct ophthalmoscope or a slit lamp biomicroscope.[15-17] With the direct ophthalmoscope, the small light spot is utilized, which is 5 degrees in size (approximately the size of the average optic disc). For direct ophthalmoscopes with 2 spot sizes, the smaller one is utilized. The middle spot is used for those with 3 spot sizes. The spot is aligned either over or adjacent to the optic disc and the size of the optic disc is compared with the size of the spot of light (*see* Figure 4). Another method consists of using a slit lamp and a high magnification fundus lens (e.g., 66 diopter [D], 78D, or 90D). A vertical slit is placed over the optic disc and the beam adjusted until it approximates the vertical disc diameter. The measurement is then read off the calibrated knob on the slit lamp. Correction factors are needed depending on the power of the fundus lens being used. A +60D lens has a correction factor of x 1.0, 78D lens x 1.1, and 90D x 1.3.[18] The average vertical optic disc diameter is 1.8 mm, and the average horizontal diameter is 1.7 mm. Finally, there are clinical situations such as in high myopia or tilted optic discs in which it is difficult to evalu-

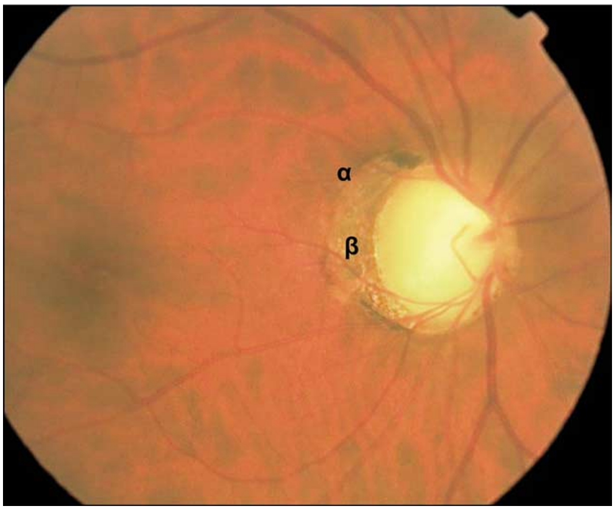

Figure 14 Parapapillary atrophy can be divided into zones α and β, with zone β being present in many eyes with glaucoma. Zone β is larger in this case and closer to the optic disc margin.

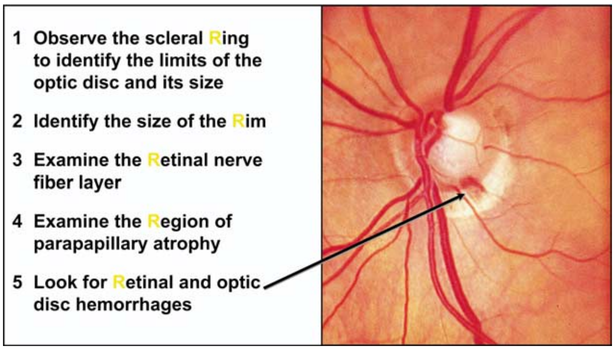

Figure 15 The fifth rule for the assessment of the optic disc is to observe for retinal and optic disc hemorrhages.

ate optic disc size because of the imprecise definition of the scleral ring and optic disc margins (*see* Figures 5 and 6).

Rule 2 relates to the need to evaluate neuroretinal rim width (*see* Figure 7). Identification of the neuroretinal rim width in all sectors of the optic disc is of fundamental importance for detection of diffuse and localized rim loss in glaucoma. The rim width is the distance between the border of the optic disc and the position of blood vessel bending. In healthy individuals, it is usually widest inferiorly, followed by superiorly, nasally, and temporally (*see* Figure 8). The mnemonic *ISNT* helps one to remember this anatomic situation. This neuroretinal rim configuration gives rise to a cup shape that is either round or horizontally oval in healthy subjects.[14] When the rim is thinner than expected and does not obey the ISNT rule, glaucomatous damage must be suspected (*see* Figure 9). Also, the color of the rim needs to be evaluated. Pallor of the rim increases the likelihood that a nonglaucomatous optic neuropathy is present, especially when pallor is greater than cup size.

Figure 16 Optic disc hemorrhages take on many appearances, depending on how recently they have occurred. They tend to occur in areas with rim tissue present.

Figure 17 The evaluation of the optic nerve in Example 1 is the following: average disc size, ISNT rule not obeyed with the inferior and superior rim being thin, a localized RNFL defect is seen at 5 o'clock, PPA is present between 3 and 5 o'clock, no hemorrhage, glaucoma is present.

Rule 3 describes the examination of the RNFL (*see* Figure 10). This is done either at the biomicroscope using a high magnification lens or by inspecting fundus photographs. If using the slit lamp and fundus lens, magnification is reduced

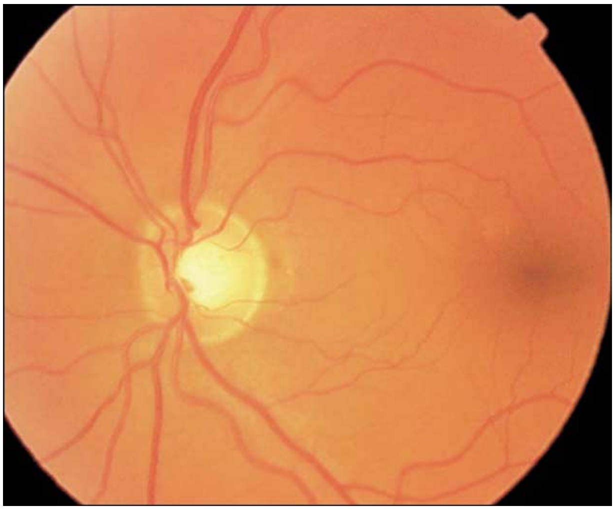

Figure 18 The evaluation of the optic nerve in Example 2 is the following: average disc size; normal rim tissue, ISNT Rule is obeyed; normal RNFL; no PPA; no hemorrhage; normal eye.

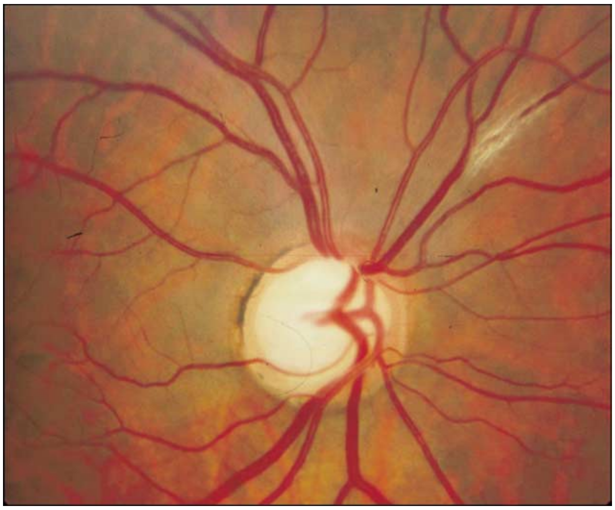

Figure 19 The evaluation of the optic nerve in Example 3 is the following: large disc size, ISNT rule not obeyed, diffuse RNFL loss, small PPA, no hemorrhage, glaucoma is present.

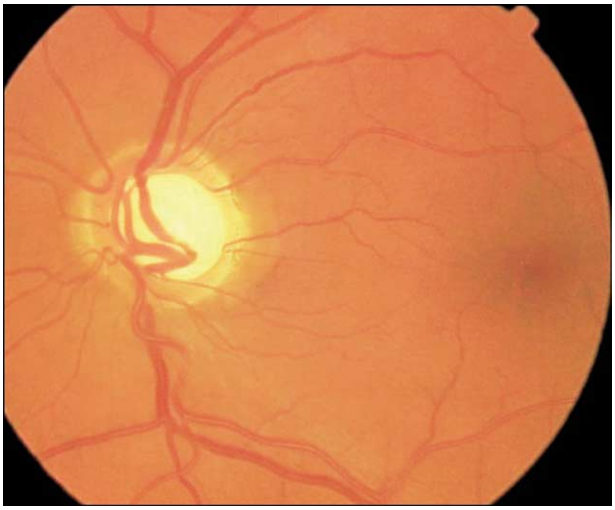

Figure 20 The evaluation of the optic nerve in Example 4 is the following: disc size is large; ISNT rule obeyed, normal rim tissue; RNFL normal; no significant PPA; only zone alpha present; no hemorrhage; normal eye with large disc.

to 6 to 10x while using a 78D or 90D lens and red-free or green light.[19] RNFL defects may also be visible in white light. In a healthy eye, bright striations are visible, and the retina glistens in the regions in which the RNFL is thickest, superior temporal and inferior temporal from the disc.[20-22] The examiner should observe the brightness and striations of the RNFL as well as the visibility of the parapapillary vessels. Typically, these vessels are indistinct but their borders become clearer as the RNFL is lost. RNFL loss can occur in a diffuse, localized, or mixed pattern.[23] With diffuse loss, there is general reduction of the RNFL brightness, with reduction of the difference normally occurring between the superior and inferior poles when compared with the temporal and nasal regions. Localized RNFL loss appears as wedge-shaped dark areas emanating from the optic disc (*see* Figure 11). These defects follow an arcuate pattern as would be expected from the normal RNFL anatomy (*see* Figure 12). True RNFL defects are at least an arteriole in width and extend back to the optic disc compared with pseudodefects, which may be thin or never extend to the optic nerve.

Rule 4 describes the examination of the parapapillary region, which is the area located just outside the optic disc (*see* Figure 13). Parapapillary atrophy (PPA) refers to the thinning and degeneration of the chorioretinal tissue just outside of the optic disc, which has an association with development and progression of glaucoma.[24-26] There are 2 forms of PPA, zone alpha (α) and zone beta (β) (*see* Figure 14). Zone α is present in most normal eyes as well as in eyes with glaucoma and is characterized by a region of irregular hypopigmentation and hyperpigmentation of the retinal pigment epithelium (RPE).[27,28] The more important zone with regard to glaucoma is zone β, which is caused by atrophy of the retinal pigment epithelium (RPE) and choriocapillaris leading to the large choroidal vessels and sclera becoming visible.[27,28] If both areas are present, zone α is always peripheral to the zone β. Zone β is more common and extensive in eyes with glaucoma than in healthy eyes. The area of PPA is spatially correlated with the area of neuroretinal rim loss, with the atrophy being largest in the corresponding area of thinner neuroretinal rim.[27-29]

Rule 5 refers to observation of retinal and optic disc hemorrhages, which are indicative of glaucoma progression (*see* Figure 15).[30-34] The hemorrhages draw their feathery shape from being present in the nerve fiber layer (*see* Figure 16). They are transient, lasting from 2 to 6 months, but may recur.[35] They may be either on the retina, be just off the disc, or bisect the disc margin and retina. They may also be located at the level of the lamina cribrosa where they tend to assume a round aspect. Sometimes the bleeding is minimal or occurs near blood vessels making detection difficult unless a meticulous examination is carried out. Hemorrhages most often are located in the inferior temporal or superior temporal regions, although they can occur in any sector of the optic disc.[36] Their color depends upon how long they have been present. They tend to be located in association with notching of the neuroretinal rim or retinal nerve fiber layer defects and may precede the later changes.[37] Although more common in normal tension glaucoma, they can be present in all types and indicate that the condition is not stable and needs further evaluation.

Several cases are provided to allow the reader to go through the 5 Rs checklist and determine whether glaucoma is present (*see* Figures 17, 18, 19, and 20). Each example illustrates a different way glaucoma may present.

Conclusion

Optic disc and RNFL assessment can be performed according to 5 rules that include the evaluation of optic disc size, rim shape and area, presence of RNFL loss, PPA, and retinal or optic disc hemorrhages. By following these 5 rules, a thorough and systematic review of the optic disc and RNFL will occur. This will improve the ability to diagnosis and manage glaucoma.

Acknowledgments

The "5 Rs" were developed by Robert N. Weinreb, M.D., and Felipe A. Medeiros, M.D. from the Hamilton Glaucoma Center, University of California, San Diego, and Remo Susanna, Jr., M.D. from the University of São Paulo. This approach was first described in a PowerPoint monograph entitled FORGE (Focusing Ophthalmology on Reframing Glaucoma Evaluation) distributed by Allergan, Inc. Figures 1, 7, 10, 13, and 15 were originally published in the FORGE™ monograph distributed by Allergan, Inc.

References

1. Weinreb RN, Khaw PT. Primary open-angle glaucoma. *Lancet* 2004;363:1711-20.
2. Medeiros FA, Weinreb RN. Medical backgrounders: glaucoma. *Drugs Today (Barc)* 2002;38:563-70.
3. Quigley HA, Dunkelberger GR, Green WR. Retinal ganglion cell atrophy correlated with automated perimetry in human eyes with glaucoma. *Am J Ophthalmol* 1989;107:453-64.
4. Sommer A, Katz J, Quigley HA, et al. Clinically detectable nerve fiber atrophy precedes the onset of glaucomatous field loss. *Arch Ophthalmol* 1991;109:77-83.
5. Harwerth RS, Carter-Dawson L, Smith EL 3rd, Barnes G, Holt WF, Crawford ML. Neural losses correlated with visual losses in clinical perimetry. *Invest Ophthalmol Vis Sci* 2004;45:3152-60.
6. Medeiros FA, Sample PA, Zangwill LM, Bowd C, Aihara M, Weinreb RN. Corneal thickness as a risk factor for visual field loss in patients with preperimetric glaucomatous optic neuropathy. *Am J Ophthalmol* 2003;136:805-13.
7. Fremont AM, Lee PP, Mangione CM, et al. Patterns of care for open-angle glaucoma in managed care. *Arch Ophthalmol* 2003;121:777-83.
8. Hertzog LH, Albrecht KG, LaBree L, et al. Glaucoma care and conformance with preferred practice patterns. Examination of the private, community-based ophthalmologist. *Ophthalmology* 1996;103:1009-13.
9. Broadway DC, Nicolela MT, Drance SM, et al. Optic disc appearance in primary open angle glaucoma. *Surv Ophthalmol* 1999;43:223-43.
10. Tuulonen A, Airaksinen P. Initial glaucomatous optic disc and retinal nerve fiber layer abnormalities and their progression. *Am J Ophthalmol* 1991;111:485-90.
11. Healey PR, Mitchell P, Smith W, et al. Relationship between cup-disc ratio and optic disc diameter: the Blue Mountains Eye Study. *Aust N Z J Ophthalmol* 1997;25 Suppl 1:S99-101.
12. Jonas JB, Zach FM, Gusek GC, et al. Pseudoglaucomatous physiologic large cups. *Am J Ophthalmol* 1989;107:137-44.
13. Garway-Heath DF, Ruben ST, Viswanathan A, et al. Vertical cup/disc ratio in relation to optic disc size: its value in the assessment of the glaucoma suspect. *Br J Ophthalmol* 1998;82:1118-24.
14. Jonas JB, Gusek GC, Naumann GO. Optic disc, cup and neuroretinal rim size, configuration and correlations in normal eyes. *Invest Ophthalmol Vis Sci* 1988;29:1151-8.
15. Jonas JB, Papastathopoulos K. Ophthalmoscopic measurement of the optic disc. *Ophthalmology* 1995;102:1102-6.
16. Garway-Heath DF, Rudnicka AR, Lowe T, et al. Measurement of optic disc size: equivalence of methods to correct for ocular magnification. *Br J Ophthalmol* 1998;82:643-9.
17. Lim CS, O'Brien C, Bolton NM. A simple clinical method to measure the optic disc size in glaucoma. *J Glaucoma* 1996;5:241-5.
18. Ansari-Shahrezaei S, Maar N, Biowski R, et al. Biomicroscopic measurement of the optic disc with a high-power positive lens. *Invest Ophthalmol Vis Sci* 2001;42:153-7.

19. Quigley HA, Sommer A. How to use nerve fiber layer examination in the management of glaucoma. *Trans Am Ophthalmol Soc* 1987;85:254-72.
20. Dichtl A, Jonas JB, Naumann GO. Retinal nerve fiber layer thickness in human eyes. *Graefes Arch Clin Exp Ophthalmol* 1999;237:474-9.
21. Jonas JB, Dichtl A. Evaluation of the retinal nerve fiber layer. *Surv Ophthalmol* 1996;40:369-78.
22. Jonas JB, Nguyen NX, Naumann GO. The retinal nerve fiber layer in normal eyes. *Ophthalmology* 1989;96:627-32.
23. Jonas JB, Budde WM, Panda-Jonas S. Ophthalmoscopic evaluation of the optic nerve head. *Surv Ophthalmol* 1999;43:293-320.
24. Jonas JB, Martus P, Horn FK, et al. Predictive factors of the optic nerve head for development or progression of glaucomatous visual field loss. *Invest Ophthalmol Vis Sci* 2004;45:2613-8.
25. Tezel G, Kolker AE, Wax MB, et al. Parapapillary chorioretinal atrophy in patients with ocular hypertension. II. An evaluation of progressive changes. *Arch Ophthalmol* 1997;115:1509-14.
26. Tezel G, Kolker AE, Kass MA, et al. Parapapillary chorioretinal atrophy in patients with ocular hypertension. I. An evaluation as a predictive factor for the development of glaucomatous damage. *Arch Ophthalmol* 1997;115:1503-8.
27. Jonas JB, Naumann GO. Parapapillary chorioretinal atrophy in normal and glaucoma eyes. II. Correlations. *Invest Ophthalmol Vis Sci* 1989;30:919-26.
28. Jonas JB, Nguyen XN, Gusek GC, et al. Parapapillary chorioretinal atrophy in normal and glaucoma eyes. I. Morphometric data. *Invest Ophthalmol Vis Sci* 1989;30:908-18.
29. Tezel G, Kass MA, Kolker AE, et al. Comparative optic disc analysis in normal pressure glaucoma, primary open-angle glaucoma, and ocular hypertension. *Ophthalmology* 1996;103:2105-13.
30. Drance S, Anderson DR, Schulzer M. Risk factors for progression of visual field abnormalities in normal-tension glaucoma. *Am J Ophthalmol* 2001;131:699-708.
31. Sonnsjo B, Dokmo Y, Krakau T. Disc haemorrhages, precursors of open angle glaucoma. *Prog Retin Eye Res* 2002;21:35-56.
32. Ishida K, Yamamoto T, Sugiyama K, et al. Disk hemorrhage is a significantly negative prognostic factor in normal-tension glaucoma. *Am J Ophthalmol* 2000;129:707-14.
33. Rasker MT, van den Enden A, Bakker D, et al. Deterioration of visual fields in patients with glaucoma with and without optic disc hemorrhages. *Arch Ophthalmol* 1997;115:1257-62.
34. Drance SM. Disc hemorrhages in the glaucomas. *Surv Ophthalmol* 1989;33:331-7.
35. Kitazawa Y, Shirato S, Yamamoto T. Optic disc hemorrhage in low-tension glaucoma. *Ophthalmology* 1986;93:853-7.
36. Jonas JB, Xu L. Optic disk hemorrhages in glaucoma. *Am J Ophthalmol* 1994;118:1-8.
37. Sugiyama K, Uchida H, Tomita G, et al. Localized wedge-shaped defects of retinal nerve fiber layer and disc hemorrhage in glaucoma. *Ophthalmology* 1999;106:1762-7.

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